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Open Geospatial Consortium

NWSVIZ EDR SERVICE PROFILE

DRAFT STANDARD
Implementation

APPROVED

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CONTENTS

I. ABSTRACT iv

II. KEYWORDS iv

III. PREFACE v

IV. SECURITY CONSIDERATIONS vi

V. SUBMITTING ORGANIZATIONS vii

VI. SUBMITTERS vii

1. SCOPE 2

2. CONFORMANCE 4

3. NORMATIVE REFERENCES 6

4. TERMS AND DEFINITIONS 8

5. REQUIREMENTS 10

ANNEX A ABSTRACT TEST SUITE 15

I ABSTRACT

This document defines the NWSViz EDR Service Profile, an OGC API — Environmental Data Retrieval (EDR) Part 3 Service Profile. It specifies normative requirements and conformance tests for server implementations conforming to this profile.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC API, EDR, Environmental Data Retrieval, NWSViz, service profile, weather



PREFACE

This document was prepared by NOAA/NWS/MDL.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- NOAA/NWS/MDL

VI SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

Table — Submitters

NAME	AFFILIATION
Shane Mill *(editor)*	NOAA/NWS/MDL

1 SCOPE

1 SCOPE

This standard defines the NWSViz EDR Service Profile. It specifies requirements and conformance tests for implementations of this profile.

2 CONFORMANCE

2 CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/nwsviz

3 NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0.

OGC API — EDR Part 3: Service Profiles (draft)

4 TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.

5 REQUIREMENTS

5 REQUIREMENTS

NOTE: Conformance with this profile is assessed against its requirements ("shall") only. Recommendations ("should") and permissions ("may") are provided as informative guidance: they are not conformance-tested and an implementation is not made non-conformant by not following them. See the quality-assessment summary emitted by the profile tooling for a count of the recommendations and permissions defined by this profile.

REQUIREMENTS CLASS 1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
CONFORMANCE CLASS	Conformance class A.1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/nwsviz`
TARGET-TYPE	NWSViz EDR Service Profile Profile Standard
NORMATIVE STATEMENTS	Requirement 1: `/req/nwsviz/conformance` Requirement 2: `/req/nwsviz/collections-endpoint` Requirement 3: `/req/nwsviz/position-coveragejson` Requirement 4: `/req/nwsviz/items-zarr-endpoint` Requirement 5: `/req/nwsviz/process-zarr-difference` Requirement 6: `/req/nwsviz/process-zarr-difference-streaming` Requirement 7: `/req/nwsviz/process-zarr-wind-streaming`

REQUIREMENT 1	
IDENTIFIER	`/req/nwsviz/conformance`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
STATEMENT	The service SHALL declare conformance to OGC API Common, EDR, EDR Part 3, and OGC API Processes.
A	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-common-1/1.0/conf/core.
B	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-common-1/1.0/conf/json.
C	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-common-1/1.0/conf/oas30.
D	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-common-2/1.0/conf/collections.

REQUIREMENT 1	
E	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-edr-1/1.0/conf/core.
F	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/conf-class-core.
G	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-processes-1/1.0/conf/core.

REQUIREMENT 2	
IDENTIFIER	`/req/nwsviz/collections-endpoint`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
STATEMENT	The service SHALL provide a /collections endpoint listing all NWSViz collections.
A	The /collections endpoint SHALL return metadata for all 13 NWSViz collections.
B	Each collection SHALL include extent, data_queries, and output_formats.

REQUIREMENT 3	
IDENTIFIER	`/req/nwsviz/position-coveragejson`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
STATEMENT	The position query SHALL return CoverageJSON.
A	Each collection SHALL provide a /collections/{id}/position endpoint.
B	The position endpoint SHALL accept a coords parameter containing a WKT POINT geometry.
C	The position endpoint SHALL return a valid CoverageJSON Coverage object as its default response.
D	The response Content-Type SHALL be application/prs.coverage+json.

REQUIREMENT 4

IDENTIFIER	`/req/nwsviz/items-zarr-endpoint`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
STATEMENT	The items query SHALL expose a virtual Zarr endpoint for each collection.
A	Each collection SHALL provide a /collections/{id}/items endpoint.
B	The items endpoint SHALL include an item with id zarr that acts as a virtual Zarr store.
C	The zarr item SHALL expose the underlying dataset as a Zarr store accessible via the .zarr/ path convention.
D	The zarr item SHALL be navigable using standard Zarr clients (e.g. xarray.open_zarr, zarr.open).
E	The zarr item response SHALL include a link with rel=zarr pointing to the root of the virtual Zarr store.
F	The virtual Zarr store SHALL reflect the full spatio-temporal extent of the collection.

REQUIREMENT 5

IDENTIFIER	`/req/nwsviz/process-zarr-difference`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
STATEMENT	The service SHALL provide the edr-zarr-difference process for computing the difference between two EDR Zarr datasets.
A	The service SHALL expose a /processes/edr-zarr-difference endpoint.
B	The process SHALL accept collection_a, instance_a, collection_b, instance_b, parameter_name, zoom_level, crs, and unit as required inputs.
C	The process SHALL accept datetime as an optional input for valid time selection.
D	The process SHALL support sync-execute job control.
E	The process SHALL return a difference_result output containing the result of (A − B) as a virtual Zarr dataset.

REQUIREMENT 6	
IDENTIFIER	`/req/nwsviz/process-zarr-difference-streaming`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
STATEMENT	The service SHALL provide the edr-zarr-difference-streaming process for streaming difference computation via a virtual Zarr endpoint.
A	The service SHALL expose a /processes/edr-zarr-difference-streaming endpoint.
B	The process SHALL accept collection_a, instance_a, collection_b, instance_b, parameter_name, and zoom_level as required inputs.
C	The process SHALL accept datetime, epsg, and units as optional inputs.
D	The process SHALL support sync-execute job control.
E	The process SHALL return a zarr_endpoint output containing a URL to stream difference chunks on-demand.

REQUIREMENT 7	
IDENTIFIER	`/req/nwsviz/process-zarr-wind-streaming`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
STATEMENT	The service SHALL provide the edr-zarr-wind-streaming process for deriving wind speed and direction from u/v components via a virtual Zarr endpoint.
A	The service SHALL expose a /processes/edr-zarr-wind-streaming endpoint.
B	The process SHALL accept collection, instance, output_parameters, and zoom_level as required inputs.
C	The output_parameters input SHALL accept wind_speed, wind_direction, or both.
D	The process SHALL accept datetime, epsg, and units as optional inputs.
E	The process SHALL support sync-execute job control.
F	The process SHALL return a zarr_endpoint output containing a URL to stream wind data chunks on-demand.

A

ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

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CONFORMANCE CLASS A.1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/nwsviz`
REQUIREMENTS CLASS	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/nwsviz`
CONFORMANCE TESTS	Abstract test A.1-1: `/conf/nwsviz/conformance` Abstract test A.1-2: `/conf/nwsviz/collections-endpoint` Abstract test A.1-3: `/conf/nwsviz/position-coveragejson` Abstract test A.1-4: `/conf/nwsviz/items-zarr-endpoint` Abstract test A.1-5: `/conf/nwsviz/process-zarr-difference` Abstract test A.1-6: `/conf/nwsviz/process-zarr-difference-streaming` Abstract test A.1-7: `/conf/nwsviz/process-zarr-wind-streaming`

Example 1

label	/conf/nwsviz/conformance
subject	/req/nwsviz/conformance
test-purpose	Validate that conformance is correctly implemented.

Send GET request to /conformance.

Verify the conformsTo array includes http://www.opengis.net/spec/ogcapi-common-1/1.0/conf/core.

Verify the conformsTo array includes http://www.opengis.net/spec/ogcapi-edr-1/1.0/conf/core.

Verify the conformsTo array includes http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/conf-class-core.

Verify the conformsTo array includes http://www.opengis.net/spec/ogcapi-processes-1/1.0/conf/core.

Example 2

label	/conf/nwsviz/collections-endpoint
subject	/req/nwsviz/collections-endpoint
test-purpose	Validate that collections endpoint is correctly implemented.

Send GET request to /collections.

Verify all 13 NWSViz collection ids are present in the response.

Verify each collection includes extent, data_queries, and output_formats.

Example 3

label	/conf/nwsviz/position-coveragejson
subject	/req/nwsviz/position-coveragejson
test-purpose	Validate that position coveragejson is correctly implemented.

Send GET request to /collections/{id}/position?coords=POINT(lon lat) for a valid coordinate within the collection extent.

Verify the response status is 200.

Verify the response Content-Type is application/prs.coverage+json.

Verify the response body is a valid CoverageJSON Coverage object with a type field equal to Coverage.

Verify the response domain contains the requested position.

Example 4

label	/conf/nwsviz/items-zarr-endpoint
subject	/req/nwsviz/items-zarr-endpoint
test-purpose	Validate that items zarr endpoint is correctly implemented.

Send GET request to /collections/{id}/items.

Verify the response includes an item with id zarr.

Verify the zarr item includes a link with rel=zarr.

Send GET request to the zarr store root (.zarr/.zmetadata or .zarr/.zattrs).

Verify the response is valid Zarr metadata JSON.

Open the zarr store using xarray.open_zarr and verify the dataset dimensions and variables are accessible.

Verify the dataset spatio-temporal extent matches the collection extent.

Example 5

label	/conf/nwsviz/process-zarr-difference
subject	/req/nwsviz/process-zarr-difference
test-purpose	Validate that process zarr difference is correctly implemented.

Send GET request to /processes/edr-zarr-difference.

Verify the process description includes inputs collection_a, instance_a, collection_b, instance_b, parameter_name, zoom_level, crs, and unit.

POST a valid execution request to /processes/edr-zarr-difference/execution with two valid collection/instance pairs.

Verify the response status is 200.

Verify the response contains a difference_result output.

Example 6

label	/conf/nwsviz/process-zarr-difference-streaming
subject	/req/nwsviz/process-zarr-difference-streaming
test-purpose	Validate that process zarr difference streaming is correctly implemented.

Send GET request to /processes/edr-zarr-difference-streaming.

Verify the process description includes inputs collection_a, instance_a, collection_b, instance_b, parameter_name, and zoom_level.

POST a valid execution request to /processes/edr-zarr-difference-streaming/execution.

Verify the response contains a zarr_endpoint output with a valid URL.

Open the returned zarr_endpoint using xarray.open_zarr and verify the difference data is accessible.

Example 7

label	/conf/nwsviz/process-zarr-wind-streaming
subject	/req/nwsviz/process-zarr-wind-streaming
test-purpose	Validate that process zarr wind streaming is correctly implemented.

Send GET request to /processes/edr-zarr-wind-streaming.

Verify the process description includes inputs collection, instance, output_parameters, and zoom_level.

POST a valid execution request with output_parameters set to [wind_speed, wind_direction].

Verify the response contains a zarr_endpoint output with a valid URL.

Open the returned zarr_endpoint using xarray.open_zarr and verify wind_speed and wind_direction variables are present.